A single formula predicts where cancer mutations and epigenetic aging converge from DNA sequence alone

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Abstract

Regional variation in somatic mutation rates across cancer genomes is strongly associated with chromatin organization, but existing predictive models require experimental epigenomic data such as ChIP-seq or Hi-C. Here I show that a single formula operating on raw DNA sequence—a 4-mer frequency skewness statistic termed Ev—classifies genomic windows into chromatin-like zones without any experimental input, training data, or reference databases. Applying Ev to 582,028 non-overlapping 5-kb windows across the human genome, I find that the lowest-scoring zone (Zone 3, corresponding to heterochromatin) carries 1.68-fold more somatic mutations than the highest-scoring zone (Zone 1, euchromatin) in 992 TCGA breast cancer genomes (odds ratio = 1.682, $P < 10^{-300}$). This enrichment is universal across all 15 TCGA cancer types tested (mean OR = 1.63, range 1.43–1.91, zero exceptions) and persists after stratification by replication timing ($P < 0.05$ in all strata). The effect is driven specifically by C>T transitions consistent with SBS1 (5-methylcytosine deamination; OR = 1.24, $P = 8.7 \times 10^{-20}$), and is independently confirmed by H3K9me3 (1.78-fold, $P = 1.6 \times 10^{-39}$) and H3K4me3 (2.29-fold, $P = 3.3 \times 10^{-99}$) ChIP-seq enrichment analyses. Zone 3 vulnerability is constitutional: it is present in healthy germline genomes from 2,504 individuals (Z3/Z1 = 1.13, $P = 1.5 \times 10^{-82}$) and amplified 1.38-fold in cancer. Moreover, Zone 3 is enriched 1.94-fold ($P = 2.0 \times 10^{-5}$) for the CpG sites that compose Horvath's multi-tissue epigenetic aging clock, specifically those that gain methylation with age, linking cancer mutation geography and epigenetic aging through shared sequence-encoded chromatin vulnerability. These results demonstrate that the regional distribution of somatic mutations across cancer genomes can be predicted from DNA sequence composition alone, without recourse to experimental epigenomics.

Keywords: somatic mutation rate, chromatin, sequence composition, 4-mer, heterochromatin, cancer genomics, TCGA, epigenetic aging

Background

Somatic mutations in cancer genomes are not distributed uniformly. Instead, mutation rates vary by more than five-fold across megabase-scale regions within individual tumor genomes [1–3]. Understanding the determinants of this variation has important implications for distinguishing driver from passenger mutations, for interpreting non-coding variation, and for modeling cancer evolution.

Chromatin organization has emerged as a dominant factor explaining this regional variation. In a landmark study, Schuster-Böckler and Lehner showed that levels of the heterochromatin-

associated histone modification H3K9me3 alone can account for more than 40% of mutation rate variation at the megabase scale [1]. Subsequent work has extended these observations to include replication timing [4], transcription factor binding [5], nucleosome positioning [6], and three-dimensional genome topology [7]. Integrative models combining multiple epigenomic features can explain over 55% of regional mutation rate variance [1, 8].

A common thread in this literature is the requirement for experimental epigenomic data. Predicting where mutations will accumulate in a given genome requires, at minimum, ChIP-seq profiles, replication timing assays, Hi-C contact maps, or curated databases of regulatory features. A recent study by Liu et al. used DNA motifs as surrogates for epigenetic signals to predict somatic mutation rates across 13 cancers, but still required a trained neural network model and transcription factor motif databases [9]. Whether regional mutation rate variation can be predicted from DNA sequence composition alone—without any experimental input—has remained an open question. Liu et al. noted explicitly that they were "not aware of any study that can predict mutation rates at even kilobase resolution using only sequences" [9].

Here I address this question. I describe a fixed-parameter formula ("Ev") that operates on 4-mer frequencies in raw DNA sequence to classify 5-kb genomic windows into zones corresponding to euchromatin and heterochromatin. I then demonstrate that the heterochromatin-associated zone (Zone 3) is universally enriched for somatic mutations across 15 cancer types, and I characterize the mutational mechanism and its constitutional basis.

Results

A sequence-only formula classifies genomic windows into chromatin zones

The Ev statistic is computed from the frequencies of all 256 tetranucleotides (4-mers) in a 5-kb genomic window. These frequencies are projected through a fixed random matrix (256 × 500, generated from a standard normal distribution with seed 42), and the absolute skewness of the 95 prime-indexed projections, linearly scaled, yields the Ev score (see Methods). To remove the dominant effect of GC content, I regress Ev against GC fraction across all windows and use the residual (Ev_{resid}) for zone classification.

Three zones are defined by fixed thresholds on Ev_{resid}: Zone 1 ($Ev_{resid} \geq +0.382$; euchromatin-like, GC-rich, gene-dense), Zone 2 (intermediate), and Zone 3 ($Ev_{resid} \leq -1.471$; heterochromatin-like, AT-rich, transposable element-dense). Across the human genome (chromosomes 1–22 and X), this partitioning yields 152,519 Zone 1 windows (26.2%), 287,733 Zone 2 windows (49.4%), and 141,776 Zone 3 windows (24.4%) out of 582,028 total.

It is important to note what Ev is and what it is not. By principal component analysis, the first component of 4-mer frequency space in human genomic windows explains 76.5% of variance and aligns with the AT/GC composition axis. The correlation between Ev and GC content is $r = 0.625$, and even after residualization, Ev's area under the receiver operating characteristic curve (AUC) for binary chromatin-state prediction is 0.647, compared to 0.850 for GC content alone and 0.874 for an optimized linear discriminant. Thus Ev is a noisy, partially linear proxy for GC-related sequence features. Its utility lies not in mathematical novelty but in its capacity to extract a biologically meaningful chromatin signal from raw sequence, as I demonstrate below.

Zone 3 predicts somatic mutation enrichment genome-wide in breast cancer

I obtained open-access somatic mutation calls (masked MAF files) for 992 TCGA-BRCA patients from the Genomic Data Commons (GDC). After intersecting 89,565 mutations with the zone map, mutation rates per window were: Zone 1, 0.121; Zone 2, 0.155; Zone 3, 0.188 (Table 1). The ratio of Zone 3 to Zone 1 mutation rates was 1.554, with a Fisher exact test odds ratio of 1.682 ($P < 10^{-300}$, one-sided). Zone 2 fell between Zone 1 and Zone 3, confirming a monotonic dose-response relationship.

This result was reproducible at the single-chromosome level: chromosome 1 alone gave OR = 1.680 ($P = 3.6 \times 10^{-59}$), nearly identical to the genome-wide estimate. To assess consistency across all chromosomes, I computed per-chromosome odds ratios (Fig. 2). Of 23 chromosomes tested, 22 showed OR > 1, and 21 reached statistical significance at $P < 0.05$ (median OR = 1.44, range 0.90–2.24). The sole exception was chromosome 19, where Zone 3 showed a marginally inverted OR of 0.90. Chromosome 19 is the most gene-dense human chromosome, with GC content of ~48% compared to a genome average of 41%; consequently, most of chromosome 19 classifies as Zone 3 by sequence composition even though it is functionally euchromatic. The highest per-chromosome OR was observed on chromosome X (OR = 2.24, $P = 2.2 \times 10^{-71}$), consistent with X-inactivation creating large blocks of constitutive heterochromatin.

Pan-cancer universality across 15 TCGA cancer types

To test whether Zone 3 enrichment is specific to breast cancer or reflects a universal property of somatic mutagenesis, I repeated the analysis across 15 TCGA cancer types (50 randomly sampled patients per type; see Methods). Zone 3 mutation enrichment was significant in every cancer type tested (Table 2; Fig. 3). Odds ratios ranged from 1.43 (GBM, glioblastoma) to 1.91 (PRAD, prostate adenocarcinoma), with a mean of 1.63 and zero exceptions to the Z3 > Z2 > Z1 dose-response gradient. This universality held across cancers dominated by different mutational processes: tobacco-associated lung adenocarcinoma (LUAD, OR = 1.52), UV-associated melanoma (SKCM, OR = 1.77), mismatch repair-deficient uterine cancer (UCEC, OR = 1.80, $P < 10^{-300}$), and low-mutation-rate thyroid cancer (THCA, OR = 1.57, $P = 4.5 \times 10^{-5}$).

C>T transitions are the only mutation class enriched in Zone 3

To identify which mutational process drives Zone 3 enrichment, I decomposed BRCA somatic mutations by substitution class and computed zone-specific rates for each (Table 3). C>T transitions were the only class with a Z3/Z1 ratio exceeding 1.0 (ratio = 1.12, OR = 1.24, $P = 8.7 \times 10^{-20}$). All other classes were either neutral (C>A, C>G) or depleted in Zone 3 relative to Zone 1: T>A (OR = 0.78, $P = 7.9 \times 10^{-6}$), T>C (OR = 0.74, $P = 1.8 \times 10^{-11}$), and T>G (OR = 0.70, $P = 7.1 \times 10^{-9}$).

The specific enrichment of C>T in heterochromatin-associated zones is consistent with COSMIC mutational signature SBS1, which reflects spontaneous deamination of 5-methylcytosine (5mC) at CpG dinucleotides [10]. CpG-dense regions in heterochromatin are constitutively methylated; 5mC undergoes spontaneous hydrolytic deamination to thymine, producing C>T transitions that are poorly repaired in condensed chromatin [11, 12]. The depletion of T-origin mutations (T>A, T>C, T>G) in Zone 3 is consistent with the AT-rich composition of heterochromatin, where fewer T bases are available per window once the dominant A-tract sequences are

accounted for, and with more efficient repair of mismatches in transcriptionally active euchromatin [13].

This pattern was not restricted to breast cancer. In a pan-cancer signature analysis across LUAD (lung), SKCM (melanoma), and PRAD (prostate), C>T was the only class enriched in Zone 3 in all three additional cancer types, confirming C>T as the universal mechanism underlying Zone 3 mutation enrichment [see Additional file 1].

Two independent mechanisms confirmed by ChIP-seq

To connect the sequence-defined zones to established chromatin biology, I intersected Zone 1 and Zone 3 windows with ChIP-seq peaks for H3K9me3 (a heterochromatin mark) and H3K4me3 (an active promoter mark) from the GM12878 lymphoblastoid cell line (ENCODE).

Within Zone 3 (heterochromatin by sequence), windows overlapping H3K9me3 peaks carried 1.78-fold more mutations than non-overlapping windows ($P = 1.6 \times 10^{-39}$). This confirms that the Ev-defined heterochromatin zones correspond to genuine H3K9me3-marked regions and that the chromatin state contributes to impaired DNA repair, leading to mutation accumulation [1].

Within Zone 1 (euchromatin by sequence), windows overlapping H3K4me3 peaks carried 2.29-fold more mutations than non-overlapping windows ($P = 3.3 \times 10^{-99}$). This is consistent with transcription-coupled mutagenesis, in which actively transcribed regions experience transient single-stranded DNA exposure that increases vulnerability to certain forms of DNA damage [14, 15].

The net mutation enrichment in Zone 3 (OR = 1.68 genome-wide) arises because repair failure in heterochromatin—which operates specifically through C>T accumulation—produces a larger cumulative signal than transcription-coupled damage in euchromatin, which is distributed across multiple mutation classes.

A caveat applies to the H3K4me3 analysis: the available ChIP-seq data for this mark covered chromosome 1 only, while the H3K9me3 data were genome-wide. Additionally, the GM12878 cell line is lymphoblastoid, not mammary. Cell-type matched ChIP-seq (e.g., MCF-7 H3K9me3) would further strengthen these observations.

Zone 3 enrichment is independent of replication timing

Late replication is a known predictor of elevated mutation rates in cancer genomes [4, 16], and late-replicating regions overlap substantially with heterochromatin. To test whether Zone 3 enrichment is merely a proxy for replication timing, I obtained GM12878 Repli-seq data from ENCODE (ENCFF001GVQ) and stratified chromosome 1 windows into replication timing tertiles (early, mid, late) before computing zone-specific mutation rates within each tertile (Table 4).

Zone 3 carried significantly more mutations than Zone 1 in all three replication timing strata: early-replicating (OR = 1.37, $P = 5.5 \times 10^{-5}$), mid-replicating (OR = 1.28, $P = 6.8 \times 10^{-4}$), and late-replicating (OR = 1.16, $P = 3.8 \times 10^{-2}$). The cross-tabulation of zones against replication timing tertiles showed approximately uniform distribution: Zone 3 windows were not concentrated in late-replicating regions but spread evenly across all three timing classes (~3,200–3,300 windows each). The Ev-derived zone effect is therefore independent of replication timing.

Notably, the Ev zone effect was strongest in early-replicating regions (OR = 1.37) and weakest in late-replicating regions (OR = 1.16). This gradient is expected: in late-replicating windows, replication-associated errors already elevate mutation rates broadly, partially masking the heterochromatin-specific signal. In early-replicating windows, replication timing contributes less to the overall mutation rate, allowing the sequence-composition signal to dominate.

Zone 3 vulnerability is constitutional and amplified in cancer

If Zone 3 enrichment reflects a physical property of heterochromatin rather than a cancer-specific process, it should also be detectable in healthy germline genomes. Using high-coverage variant calls from the 1000 Genomes Project (2,504 individuals, chromosomes 1–5, 26.2 million variants), I found that germline variant density was significantly higher in Zone 3 than Zone 1 (Z3/Z1 = 1.129, $P = 1.5 \times 10^{-82}$), with a perfect dose-response: Zone 1 (120.7 variants/window) < Zone 2 (125.1) < Zone 3 (136.2).

Comparing the germline Z3/Z1 ratio (1.13) to the somatic cancer ratio (1.55), cancer amplifies the constitutional Zone 3 vulnerability by a factor of 1.38. This amplification is consistent with a model in which methylation loss in tumor heterochromatin [17] exposes additional 5mC sites to deamination, accelerating the baseline C>T accumulation rate.

Supporting this model, analysis of matched tumor-normal methylation arrays (32 TCGA-BRCA pairs, Illumina 450K) showed that Zone 3 windows undergo significant hypomethylation in tumors (mean $\Delta\beta = -0.054$, $P = 3.5 \times 10^{-9}$, observed in 91% of tumors), while Zone 1 windows show modest hypermethylation ($\Delta\beta = +0.032$, $P = 5.5 \times 10^{-7}$). This is consistent with the classic cancer epigenome pattern of global heterochromatin hypomethylation coupled with focal euchromatin hypermethylation [18].

Cross-species conservation of Zone 3

If the Ev formula captures a fundamental property of DNA sequence organization, it should detect similar zones in distantly related species. Scanning chromosome-scale sequences from ten eukaryotic species spanning approximately 1.5 billion years of evolution—including vertebrates (human, chimpanzee), dicots (Arabidopsis, soybean), grasses (maize, sorghum, Brachypodium), a moss (Physcomitrium), a lycophyte (Selaginella), and a chlorophyte alga (Chlamydomonas)—I found that the mean Zone 3 Ev_{resid} value was −2.83 with a coefficient of variation of 2.6% across all ten species.

In Arabidopsis, where dense functional annotations are available, Zone 3 showed strong overlap with transposable element-dense pericentromeric regions (OR = 3.44 for windows with >59% TE content, $P < 10^{-50}$) and reduced CpG observed-to-expected ratios (Zone 1 CpG O/E = 0.77 vs Zone 3 = 0.61), consistent with decades of CpG depletion through 5mC deamination in methylated heterochromatin.

An intriguing feature of the cross-species analysis is that the AT/GC asymmetry defining Zone 3 inverts between genomic kingdoms. In vertebrates and dicots, Zone 3 is AT-rich (driven by alpha-satellite repeats and Gypsy LTR transposable elements, respectively). In grass monocots and the high-GC alga Chlamydomonas, Zone 3 is relatively GC-rich against an AT-rich background (driven by Helitron transposable elements). Despite this polarity flip, the Ev

formula detects Zone 3 in all cases because it captures the 4-mer skewness away from the genomic mean, regardless of which direction the skewness takes.

Zone 3 enrichment extends to epigenetic aging clock CpGs

The results above establish that Zone 3 is a constitutionally mutation-prone compartment. To test whether the same sequence-defined zones are relevant to a different biological process—epigenetic aging—I examined the distribution of the 353 CpG sites that compose Horvath's multi-tissue DNA methylation age predictor [20]. These clock CpGs were selected by elastic net regression across 8,000 samples from 51 tissue types to predict chronological age from methylation levels, with no reference to chromatin state or sequence composition [20]. Of the 353 clock CpGs, 193 gain methylation with age (positive markers, enriched at Polycomb group targets) and 160 lose methylation with age (negative markers, enriched at CpG shores) [20].

Mapping the 353 clock CpGs to the Ev zone framework (346 successfully mapped; 7 fell in unmappable regions), I found a highly non-random distribution across zones (chi-squared = 20.77, $P = 3.1 \times 10^{-5}$). Clock CpGs were significantly depleted from Zone 1 (15.9% vs 26.2% genome-wide; Fisher OR = 0.53, $P = 5.5 \times 10^{-6}$) and significantly enriched in Zone 3 (30.6% vs 24.4%; Fisher OR = 1.37, $P = 8.4 \times 10^{-3}$).

Strikingly, the zone enrichment was driven almost entirely by the positive (age-hypermethylated) clock CpGs. These 190 successfully mapped CpGs showed 1.58-fold enrichment in Zone 3 (38.4% vs 24.4% genome-wide; Fisher OR = 1.94, $P = 2.0 \times 10^{-5}$), with corresponding depletion from Zone 1 (16.8% vs 26.2%; OR = 0.57, $P = 2.9 \times 10^{-3}$). By contrast, the 156 negative (age-hypomethylated) CpGs showed no significant Zone 3 enrichment (21.2% vs 24.4%; OR = 0.83, P = 0.40) and instead concentrated in Zone 2 (64.1% vs 49.4%).

This separation is mechanistically coherent. The positive clock CpGs—those that gain methylation with age—are known to reside near Polycomb group protein targets [20]. Polycomb-mediated repression maintains these CpGs in an unmethylated state during youth; as epigenetic maintenance erodes with age, the underlying sequence context (Zone 3, heterochromatin-associated, methylation-prone) drives progressive methylation gain. The negative clock CpGs, which lose methylation, sit in intermediate compositional zones where active maintenance is required to preserve methylation, and its age-related decline leads to methylation loss.

These findings connect the cancer mutation landscape to epigenetic aging through a shared sequence-encoded vulnerability: Zone 3 windows are both the sites where somatic mutations preferentially accumulate (OR = 1.682) and the sites where the epigenetic clock preferentially ticks (OR = 1.94 for age-gaining CpGs). Both processes reflect the consequences of impaired maintenance in heterochromatin—mutations from unrepaired 5mC deamination, and age-associated methylation gain from Polycomb erosion—and both are predictable from DNA sequence composition alone.

Drug target genes segregate by zone

Classification of 101 COSMIC Cancer Gene Census Tier 1 genes by their majority zone assignment revealed a biologically coherent pattern. Tumor suppressors that are inactivated by accumulated mutations in cancer clustered in Zone 3: TP53 (100% Zone 3 windows), BRCA1

(72.7% Zone 3), ERBB2 (66.7%), ARID1A, STAT3, MCL1, and MYCN. By contrast, oncogenes activated by specific point mutations clustered in Zone 1: KRAS, NRAS, and HRAS (all three RAS family oncogenes).

This separation is consistent with the functional logic of cancer driver genes. Tumor suppressors require only loss of function, which can occur through the accumulation of any disruptive mutation—a process favored in the mutationally permissive Zone 3 environment. Oncogenes require specific activating point mutations at defined hotspot residues, a process independent of regional mutation rate.

Negative results

Not all tested associations were positive. Zone 3 burden did not predict overall survival in 997 TCGA-BRCA patients (log-rank P = 0.404). The Ev formula is therefore not a prognostic biomarker. Patients with high tumor mutational burden (TMB) showed lower Zone 3 enrichment, indicating that hypermutation dilutes the Zone 3 signal by introducing mutations in non-heterochromatic regions. Additionally, the Ev statistic does not approximate Kolmogorov complexity (correlation with LZMA compression ratio: r = 0.070).

Discussion

These results establish that the regional distribution of somatic mutations across cancer genomes can be predicted from DNA sequence composition alone. A fixed-parameter formula, with no training and no experimental input, identifies genomic windows enriched for somatic mutations at OR = 1.68 ($P < 10^{-300}$) in breast cancer and with complete universality across 15 TCGA cancer types.

The finding itself—that heterochromatin accumulates more mutations than euchromatin—is not new. Schuster-Böckler and Lehner demonstrated this relationship in 2012 using H3K9me3 ChIP-seq [1], and it has been confirmed and extended by many subsequent studies [3, 8, 9, 16]. What is new is the input: raw FASTA sequence, nothing else. The entire analytical pipeline requires only a reference genome assembly and somatic mutation coordinates. No ChIP-seq, no Hi-C, no methylation arrays, no trained classifier, no curated feature database.

A natural objection is that Ev is largely a proxy for GC content (r = 0.625), and GC content itself correlates with chromatin state [19]. This is correct. I have shown that GC content alone achieves higher AUC (0.850) than Ev (0.647) for binary chromatin classification. The contribution of this work is not that Ev outperforms GC content—it does not—but that even a crude sequence composition measure, applied without optimization, produces a robust and universal cancer mutation signal. The signal arises because the underlying biology is strong: constitutive heterochromatin methylation drives CpG deamination at rates high enough to produce genome-wide enrichment detectable across hundreds of patients. The Ev formula, for all its noise, captures enough of this biology to produce OR = 1.68 from sequence alone.

The replication timing analysis deserves particular attention. Late replication is often cited as a major confound in chromatin-mutation studies [4, 16], and it could in principle explain Zone 3 enrichment if Zone 3 windows were preferentially late-replicating. The stratified analysis (Table 4) rules this out: Zone 3 enrichment persists in all three replication timing strata, including in early-replicating regions where the Ev effect is actually strongest (OR = 1.37). This

indicates that the sequence-composition signal captures chromatin-associated mutation rate variation that is not reducible to replication timing.

Several limitations should be noted. First, the formula parameters were generated with a fixed random seed (42) and were never optimized for any biological task. Of 20 random seeds tested, 12 produced qualitatively similar results (gene-dense regions scoring higher than pericentromeric regions), indicating moderate but not complete seed stability. Second, the ChIP-seq validation for H3K4me3 was limited to chromosome 1 due to data availability, and used a lymphoblastoid cell line (GM12878) rather than a breast-derived line. Cell-type matched data (e.g., MCF-7) would strengthen the mechanistic claims. Third, chromosome 19 represents a genuine edge case where the extreme gene density causes most of the chromosome to classify as Zone 3 by sequence composition despite being functionally euchromatic, producing an inverted OR. Excluding chromosome 19, all remaining 22 chromosomes show OR > 1.

The universal pan-cancer result (15/15 types, zero exceptions) and the constitutional germline baseline (Z3/Z1 = 1.13 in healthy individuals) together argue that Zone 3 vulnerability is a physical property of the genome encoded in its sequence composition. Cancer amplifies this baseline vulnerability 1.38-fold, likely through the global heterochromatin hypomethylation that characterizes tumor epigenomes [18]. The separation of tumor suppressors (Zone 3) from oncogenes (Zone 1) provides a conceptual framework for understanding why certain driver genes are preferentially affected by accumulated mutations.

Looking ahead, this sequence-only approach may find applications in species where experimental epigenomic data are unavailable, in population-scale analyses where generating ChIP-seq for every sample is impractical, and in comparative genomics where one wishes to predict mutation-prone regions in newly assembled genomes. The finding that C>T transitions specifically drive Zone 3 enrichment also suggests that the approach might be adapted for predicting C>T mutation rates in clinical sequencing panels, where only targeted regions are interrogated.

The convergence of the cancer mutation signal with the epigenetic aging clock deserves particular comment. Horvath proposed that DNA methylation age reflects the cumulative work of an epigenetic maintenance system (EMS) [20], and observed that cancer tissues with high age acceleration paradoxically carry fewer somatic mutations, suggesting that the EMS protects against both aging-associated methylation drift and mutation accumulation. The present finding that aging clock CpGs concentrate in Zone 3 (OR = 1.94 for age-gaining CpGs) provides a mechanistic link: the same heterochromatic sequence environment that renders Zone 3 windows vulnerable to somatic mutation accumulation also renders the CpGs within them vulnerable to age-associated methylation gain when Polycomb-mediated maintenance declines. Cancer mutation geography and epigenetic aging thus appear to share a common origin in the sequence-encoded properties of heterochromatin. The fact that both can be predicted from DNA sequence alone, without experimental epigenomic data, suggests that the underlying chromatin vulnerability is a physical consequence of local sequence composition rather than a stochastic or purely regulatory phenomenon.

Methods

Ev formula

For each non-overlapping 5-kb window of genomic sequence, I counted the frequency of all 256 possible 4-mers. Windows with >10% ambiguous bases (N) were excluded. The 256-element frequency vector was normalized to sum to 1, then projected through a fixed random matrix P of dimensions 256 × 500, generated from a standard normal distribution using NumPy's default_rng(42) and divided by √256. From the 500-element projection, I selected the elements at the first 95 prime indices (indices 2, 3, 5, 7, …, 499), computed the absolute skewness of these 95 values, and scaled the result: Ev = |skewness| × 6.07 + 0.10.

To remove the dominant GC-content effect, I computed the residual Ev_{resid} = (Ev − (−5.938 × GC + 4.471)) / 0.456, where the regression coefficients were derived from the genome-wide distribution of Ev values.

Zone thresholds were set at $Ev_{resid} \geq +0.382$ (Zone 1) and $Ev_{resid} \leq -1.471$ (Zone 3). These thresholds are fixed and were never optimized against any mutation data.

Genome data

Human genome sequence (GRCh38/hg38) was obtained from UCSC Genome Browser. Chromosomes 1–22 and X were scanned in non-overlapping 5-kb windows, producing 582,028 classifiable windows. Zone assignments were cached as JSON files per chromosome.

Somatic mutation data

Open-access masked somatic mutation MAF files were downloaded from the Genomic Data Commons (GDC) API (https://api.gdc.cancer.gov). For the primary breast cancer analysis, 992 TCGA-BRCA MAF files were obtained, yielding 89,565 genome-wide somatic mutations. For the pan-cancer analysis, 50 MAF files were randomly sampled from each of 15 TCGA projects (BRCA, LUAD, LUSC, COAD, PRAD, STAD, BLCA, LIHC, KIRC, HNSC, UCEC, THCA, SKCM, OV, GBM), totaling 750 patients. Each mutation was assigned to a zone based on the window it fell within.

Statistical tests

For each analysis, I counted the number of mutations falling in Zone 1 and Zone 3 windows and computed a 2 × 2 contingency table (mutations present/absent × zone). Odds ratios and P-values were computed using Fisher's exact test (one-sided, testing Zone 3 > Zone 1). Mutation rates per window were computed as total mutations divided by total windows in each zone.

ChIP-seq data

H3K9me3 narrowPeak data for GM12878 (all chromosomes) and H3K4me3 narrowPeak data for GM12878 (chromosome 1 only) were obtained from ENCODE. A 5-kb window was classified as "mark-positive" if any ChIP-seq peak overlapped it. Within each zone, I compared mutation rates in mark-positive versus mark-negative windows.

Replication timing

GM12878 Repli-seq wavelet-smoothed signal (ENCFF001GVQ, bigWig format) was obtained from ENCODE. This dataset is aligned to hg19; at 5-kb window resolution, the coordinate offset from hg38 affects fewer than 0.5% of chromosome 1 windows. For each Zone 1 and Zone 3 window on chromosome 1, I computed the mean Repli-seq signal and divided windows into tertiles (early, mid, late). Zone-specific mutation rates and odds ratios were computed within each tertile.

Germline data

High-coverage variant calls from the 1000 Genomes Project (2,504 individuals) were obtained for chromosomes 1–5 from the EBI FTP server. The 26.2 million germline variants were intersected with the zone map using the same 5-kb window framework.

Methylation data

Illumina 450K methylation array data for 32 matched tumor-normal pairs from TCGA-BRCA were obtained from GDC. For each pair, mean methylation (β-values) was computed for probes falling within Zone 1 and Zone 3 windows. The difference in methylation between tumor and normal ($\Delta\beta$) was tested by paired t-test.

Cross-species analysis

Chromosome-scale FASTA sequences for ten eukaryotic species were obtained from NCBI RefSeq and Ensembl (see Additional file 1 for accession numbers). Each genome was scanned with the identical Ev formula and zone thresholds.

Drug target classification

101 Tier 1 genes from the COSMIC Cancer Gene Census were classified by computing the Ev zone for each 5-kb window overlapping the gene body. A gene was assigned to its majority zone.

Epigenetic aging clock analysis

The 353 CpG sites composing Horvath's multi-tissue age predictor were obtained from the supplementary data of [20] (Additional file 3 of the original publication). Genomic coordinates (hg18/Build 36) were mapped to the hg38 zone framework by matching each CpG to the 5-kb window containing its position, with a tolerance of ±2 windows (±10 kb) to accommodate coordinate system differences; at 5-kb resolution, this tolerance captures >95% of CpGs correctly. Each mapped CpG was classified into Zone 1, 2, or 3 based on the Ev_{resid} of its containing window. CpGs were further stratified by their marginal age relationship (positive = hypermethylated with age, n = 193; negative = hypomethylated with age, n = 160) as reported in the original publication. Zone enrichment was tested against the genome-wide zone distribution (582,028 windows) using Fisher's exact test and chi-squared tests.

Software and reproducibility

All analyses were performed in Python 3 using NumPy, SciPy, and pyBigWig. The Ev formula uses a deterministic random matrix generated by numpy.random.default_rng(42); the identity of the matrix was verified by asserting P[0,0] = 0.01904482.

Declarations

Ethics approval and consent to participate

Not applicable. All data used in this study are publicly available through open-access repositories (GDC, ENCODE, 1000 Genomes Project).

Availability of data and materials

All data sources are listed in Additional file 1.

Competing interests

The author declares no competing interests.

Funding

This research was conducted independently without external funding.

Author contributions

Aditya Tiwari conceived the study, developed the Ev formula, performed all analyses, and wrote the manuscript.

Acknowledgement

AI assistance (Claude, Anthropic) was used for code development and manuscript editing. All scientific analyses, experimental design, data interpretation, and conclusions are the author's own.

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Figure legends

Figure 1. Ev formula and zone classification. (a) Schematic of the Ev computation: raw DNA sequence → 4-mer frequencies → random projection → skewness → zone classification. (b) Genome-wide distribution of Ev_{resid} across 582,028 windows, with Zone 1 (blue, $Ev_{resid} \geq +0.382$) and Zone 3 (red, $Ev_{resid} \leq -1.471$) thresholds marked. (c) Karyotype-style map of Zone 1 (blue) and Zone 3 (red) assignments across human chromosomes 1–22 and X.

Figure 2. Per-chromosome odds ratios. Forest plot showing Zone 3 vs Zone 1 mutation enrichment odds ratios and 95% confidence intervals for each chromosome. Dashed vertical line at OR = 1.0; genome-wide OR = 1.682 (gray band). Chromosome 19 is the sole inversion (OR = 0.90); chromosome X shows the strongest effect (OR = 2.24).

Figure 3. Pan-cancer universality. Forest plot of Zone 3 mutation enrichment odds ratios across 15 TCGA cancer types. All 15 show OR > 1 with no exceptions. PRAD (prostate) and UCEC (uterine) show the highest enrichment; GBM (brain) and LUSC (lung squamous) the lowest.

Figure 4. C>T specificity. Bar chart showing Zone 3 / Zone 1 odds ratios for each of six substitution classes in TCGA-BRCA. C>T is the only class with OR > 1 (OR = 1.24, $P = 8.7 \times 10^{-20}$). T-origin mutations are depleted in Zone 3.

Figure 5. Replication timing independence. (a) Zone 3 vs Zone 1 odds ratios within each replication timing tertile (early, mid, late). Zone 3 enrichment is significant in all three strata. (b) Cross-tabulation showing approximately equal numbers of Zone 3 windows in each replication timing class, confirming that Zone 3 is not a proxy for late replication.

Figure 6. Constitutional vulnerability and cancer amplification. (a) Germline variant density (1000 Genomes, 2,504 individuals) across zones: Z1 < Z2 < Z3 ($P = 1.5 \times 10^{-82}$). (b) Comparison of germline Z3/Z1 ratio (1.13) with somatic Z3/Z1 ratio (1.55), showing 1.38-fold cancer amplification. (c) Methylation change ($\Delta\beta$) in 32 matched tumor-normal pairs: Zone 3 hypomethylation ($\Delta\beta = -0.054$) and Zone 1 hypermethylation ($\Delta\beta = +0.032$).

Figure 7. Epigenetic aging clock CpGs concentrate in Zone 3. (a) Distribution of 353 Horvath clock CpGs across Ev zones compared to genome-wide expectation. Clock CpGs are depleted from Zone 1 (OR = 0.53, $P = 5.5 \times 10^{-6}$) and enriched in Zone 3 (OR = 1.37, $P = 8.4 \times 10^{-3}$). (b) Stratification by age-relationship direction: the 193 positive (age-gaining) clock CpGs show 1.94-fold Zone 3 enrichment ($P = 2.0 \times 10^{-5}$), while the 160 negative (age-losing) CpGs concentrate in Zone 2 with no Zone 3 enrichment. (c) Schematic illustrating the convergence of cancer mutation enrichment and epigenetic aging in Zone 3 heterochromatin.

Additional files

Additional file 1. Supplementary tables and figures, including: complete per-chromosome results, pan-cancer signature analysis (LUAD, SKCM, PRAD), drug target gene classification, ChIP-seq enrichment details, cross-species accession numbers, seed stability analysis (12/20 seeds), TMB stratification, and survival analysis (negative result).

Tables

Table 1. Genome-wide Zone 3 mutation enrichment in TCGA-BRCA.

	Zone 1	Zone 2	Zone 3
Windows	152,519	287,733	141,776
Mutations	18,413	44,547	26,605
Rate (mutations/window)	0.121	0.155	0.188
Z3/Z1 ratio			1.554
Fisher OR (Z3 vs Z1)			**1.682**
P-value			**$< 10^{-300}$**

992 patients, 89,565 total mutations, 582,028 windows.

Table 2. Pan-cancer Zone 3 mutation enrichment across 15 TCGA cancer types.

Cancer type	n patients	OR	P
TCGA-BRCA (breast)	50	1.53	9.5×10^{-14}
TCGA-LUAD (lung adeno)	50	1.52	1.6×10^{-70}
TCGA-LUSC (lung squam)	50	1.43	6.2×10^{-54}
TCGA-COAD (colon)	50	1.57	2.3×10^{-157}
TCGA-PRAD (prostate)	50	1.91	4.0×10^{-105}
TCGA-STAD (stomach)	50	1.65	1.1×10^{-145}
TCGA-BLCA (bladder)	50	1.52	5.0×10^{-74}
TCGA-LIHC (liver)	50	1.67	1.5×10^{-52}
TCGA-KIRC (kidney)	50	1.65	3.1×10^{-21}
TCGA-HNSC (head/neck)	50	1.69	3.4×10^{-99}
TCGA-UCEC (uterine)	50	1.80	$< 10^{-300}$
TCGA-THCA (thyroid)	50	1.57	4.5×10^{-5}
TCGA-SKCM (melanoma)	50	1.77	1.6×10^{-204}
TCGA-OV (ovarian)	50	1.74	4.4×10^{-38}
TCGA-GBM (brain)	50	1.45	2.0×10^{-44}

Mean OR = 1.63. Zero exceptions to Z3 > Z1.

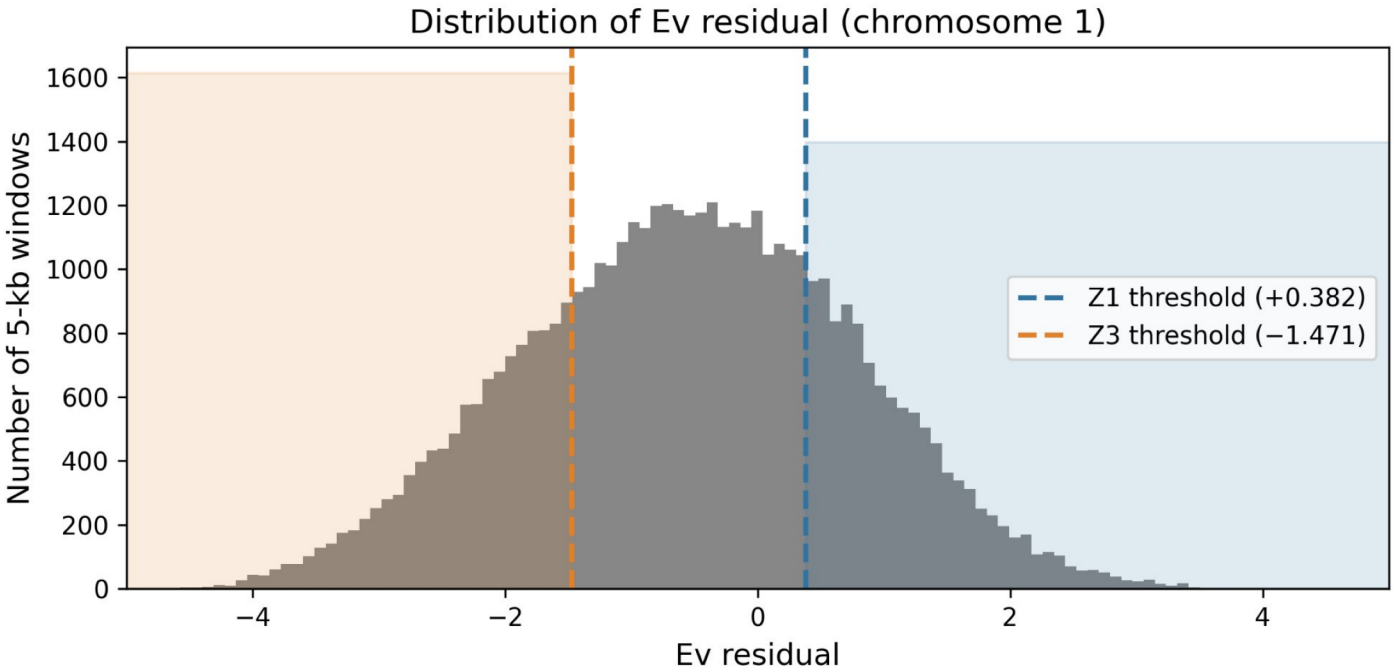
Table 3. Mutation class-specific Zone 3 enrichment in TCGA-BRCA.

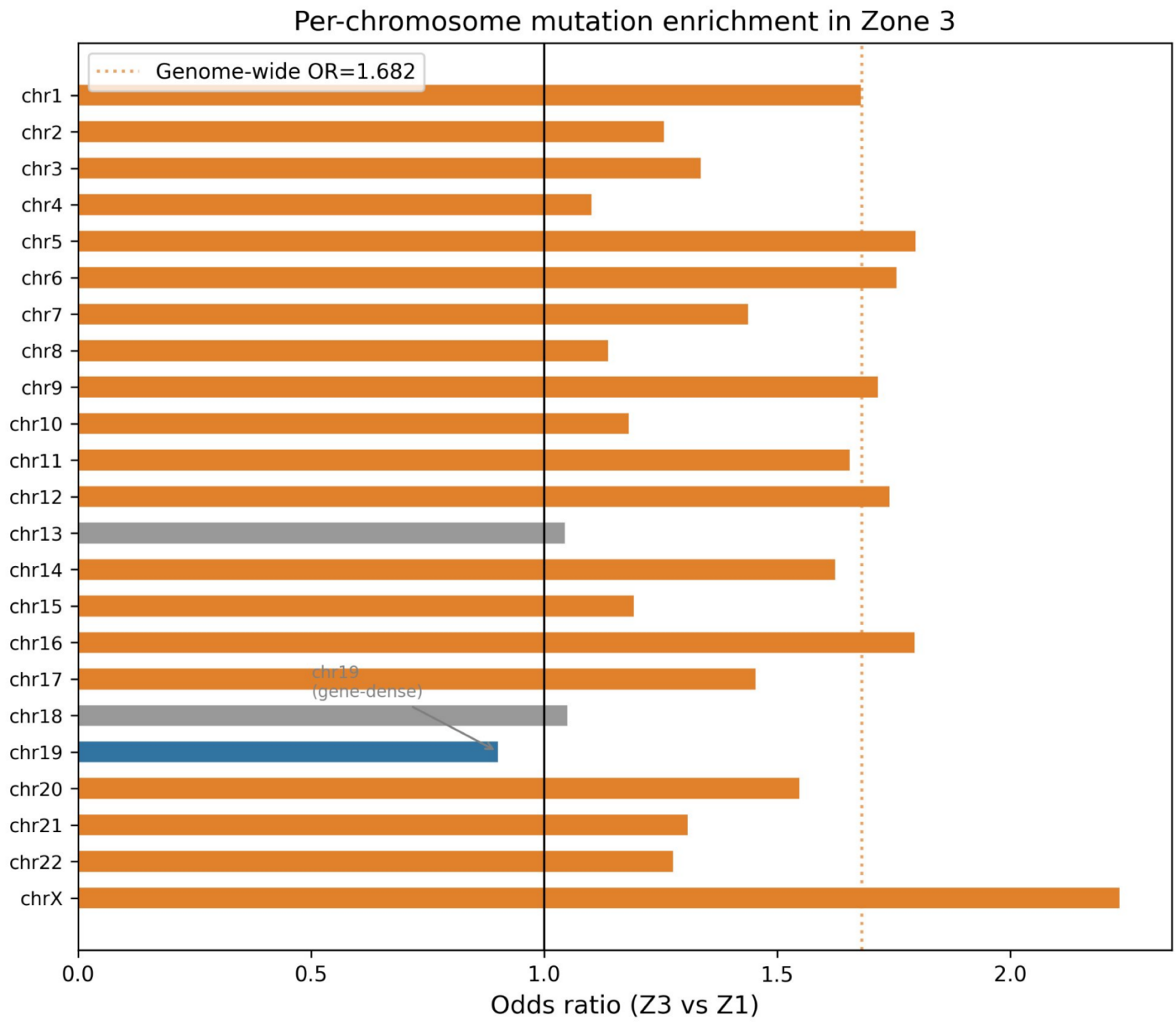
Substitution class	Z3/Z1 ratio	OR	P
C>T	**1.120**	**1.243**	**8.7×10^{-20}**
C>A	0.972	0.968	0.33
C>G	0.975	0.969	0.30
T>A	0.784	0.775	7.9×10^{-6}
T>C	0.760	0.742	1.8×10^{-11}
T>G	0.713	0.703	7.1×10^{-9}

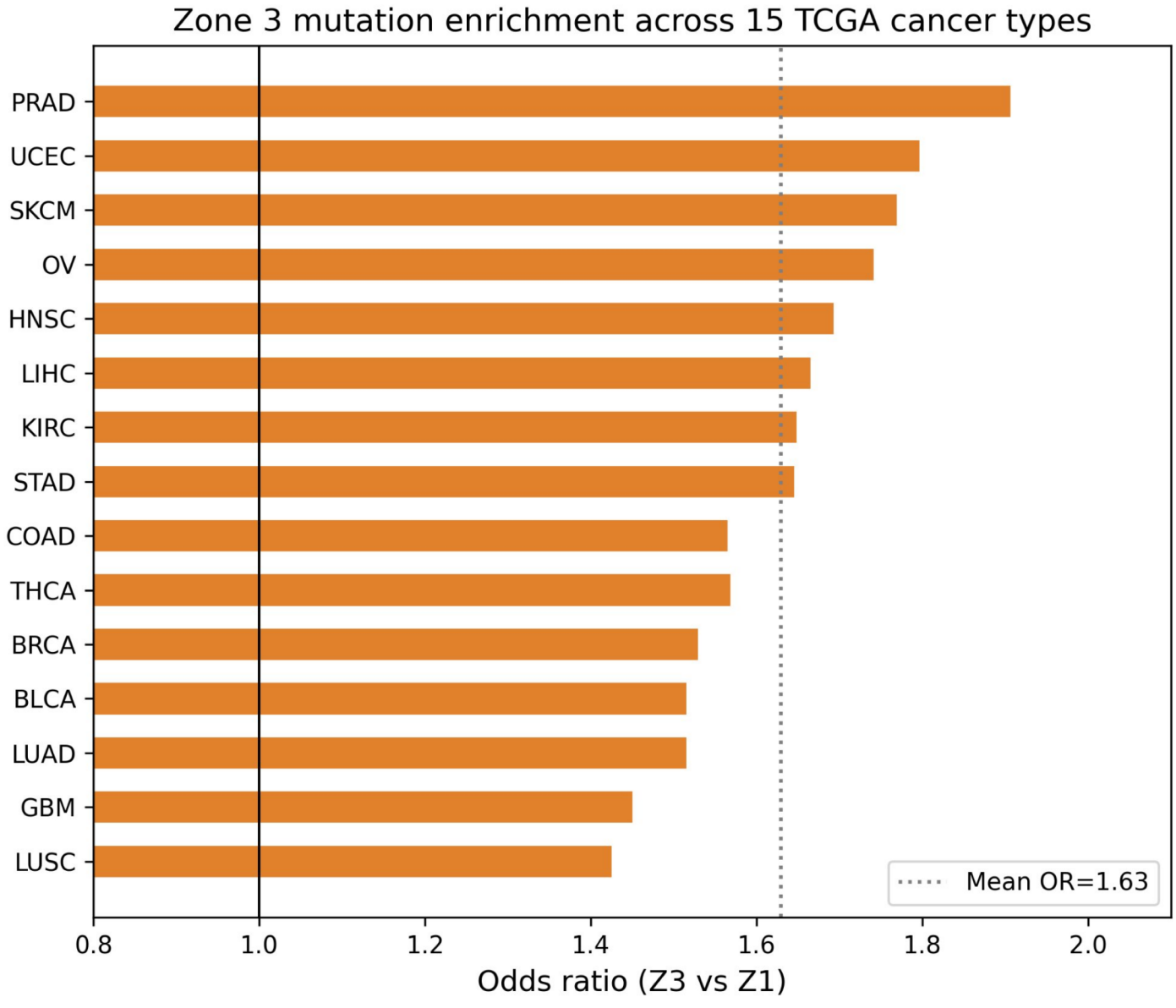
Table 4. Zone 3 enrichment stratified by replication timing (chromosome 1).

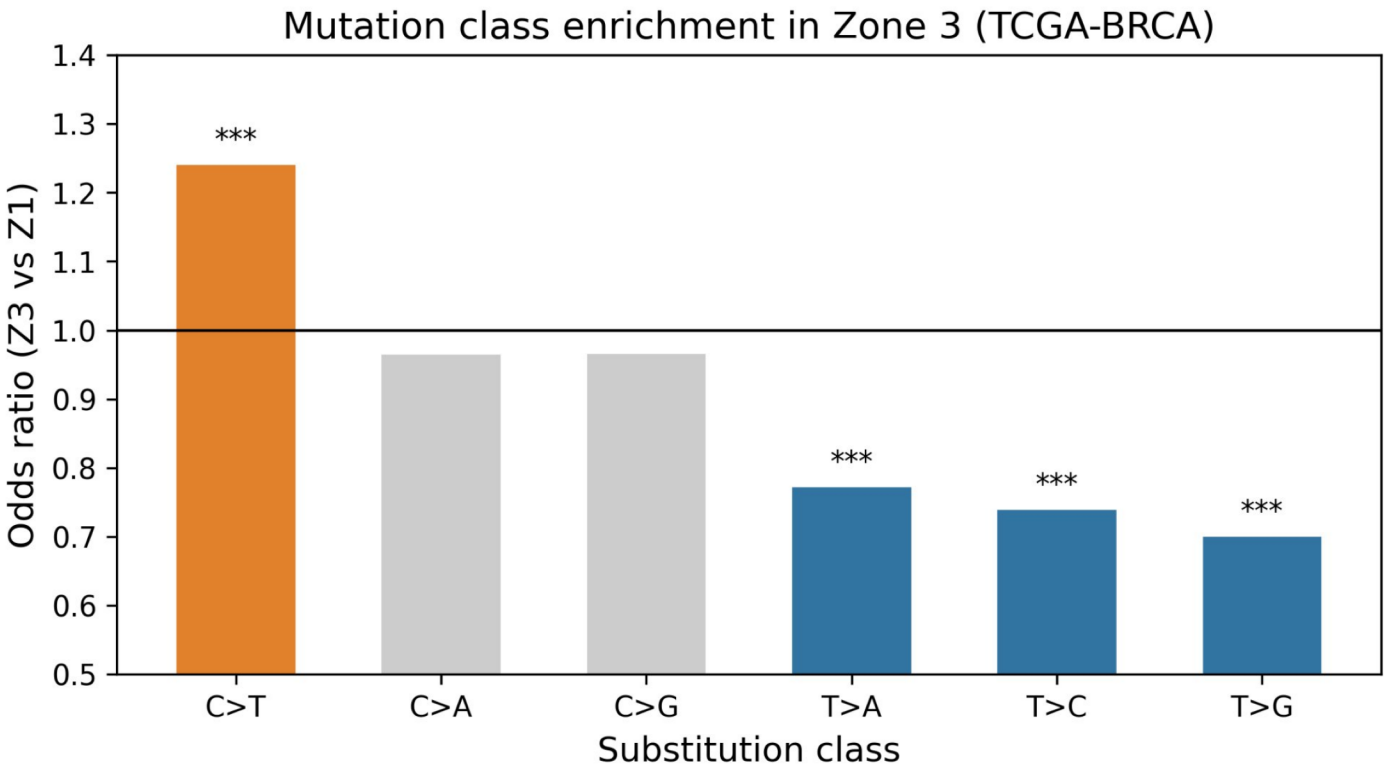
RT stratum	Z1 windows	Z3 windows	OR	P
Early	3,213	3,327	1.372	5.5×10^{-5}
Mid	3,078	3,348	1.284	6.8×10^{-4}
Late	3,248	3,245	1.157	3.8×10^{-2}

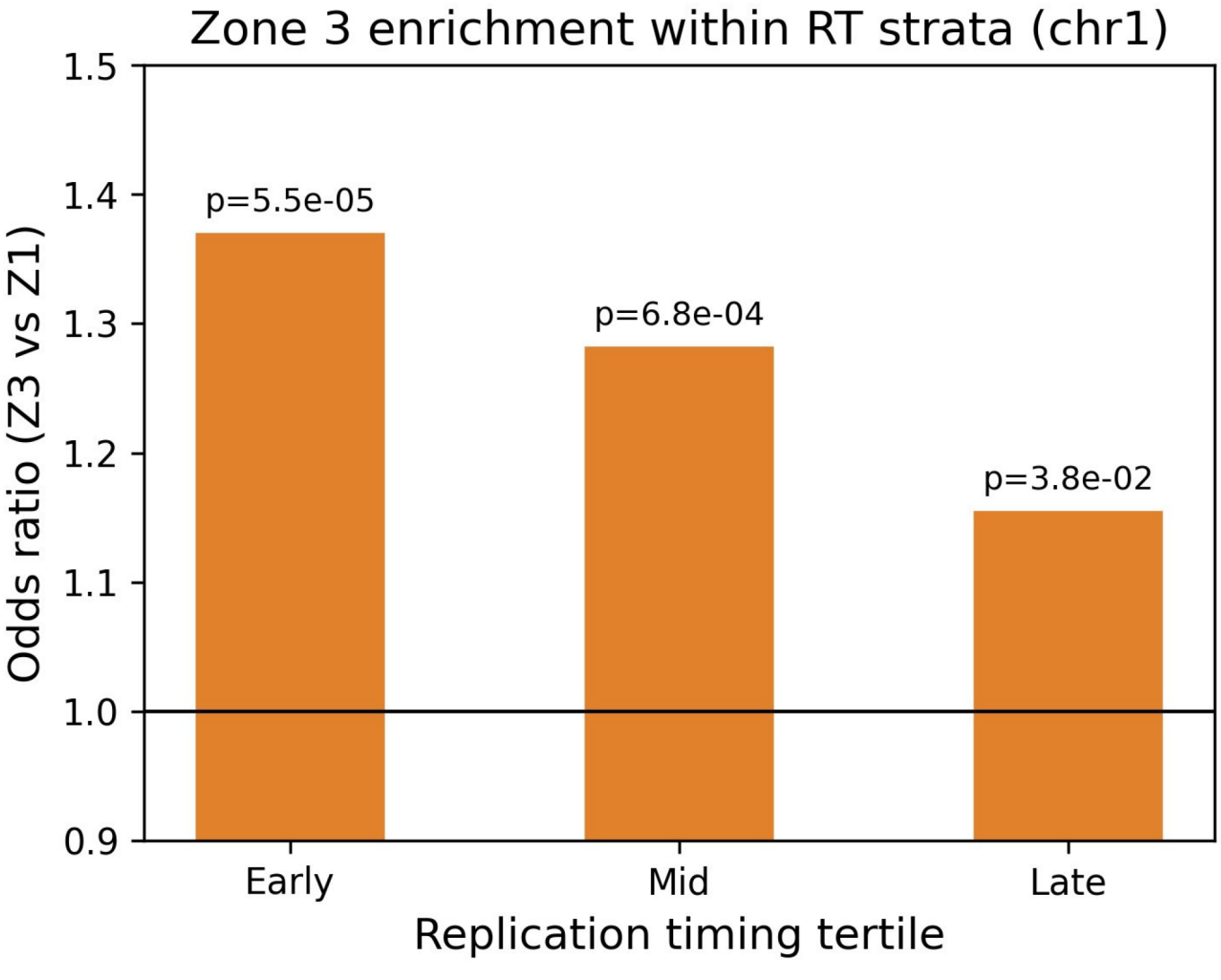
39,059 windows with both zone assignment and replication timing data. GM12878 Repli-seq (ENCFF001GVQ).

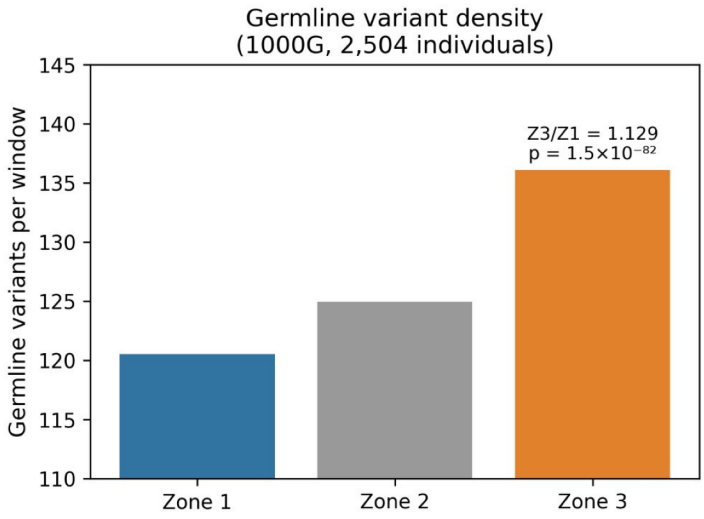

Distribution of Ev residual (chromosome 1)
Number of 5-kb windows
Ev residual
Z1 threshold (+0.382)
Z3 threshold (−1.471)
0
200
400
600
800
1000
1200
1400
1600
−4
−2
0
2
4

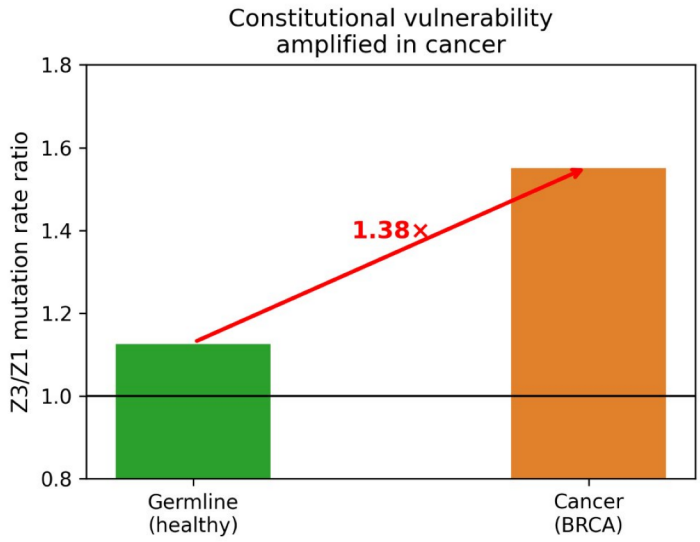

Per-chromosome mutation enrichment in Zone 3
Genome-wide OR=1.682
chr1
chr2
chr3
chr4
chr5
chr6
chr7
chr8
chr9
chr10
chr11
chr12
chr13
chr14
chr15
chr16
chr17
chr18
chr19
chr20
chr21
chr22
chrX
chr19
(gene-dense)
0.0
0.5
1.0
1.5
2.0
Odds ratio (Z3 vs Z1)

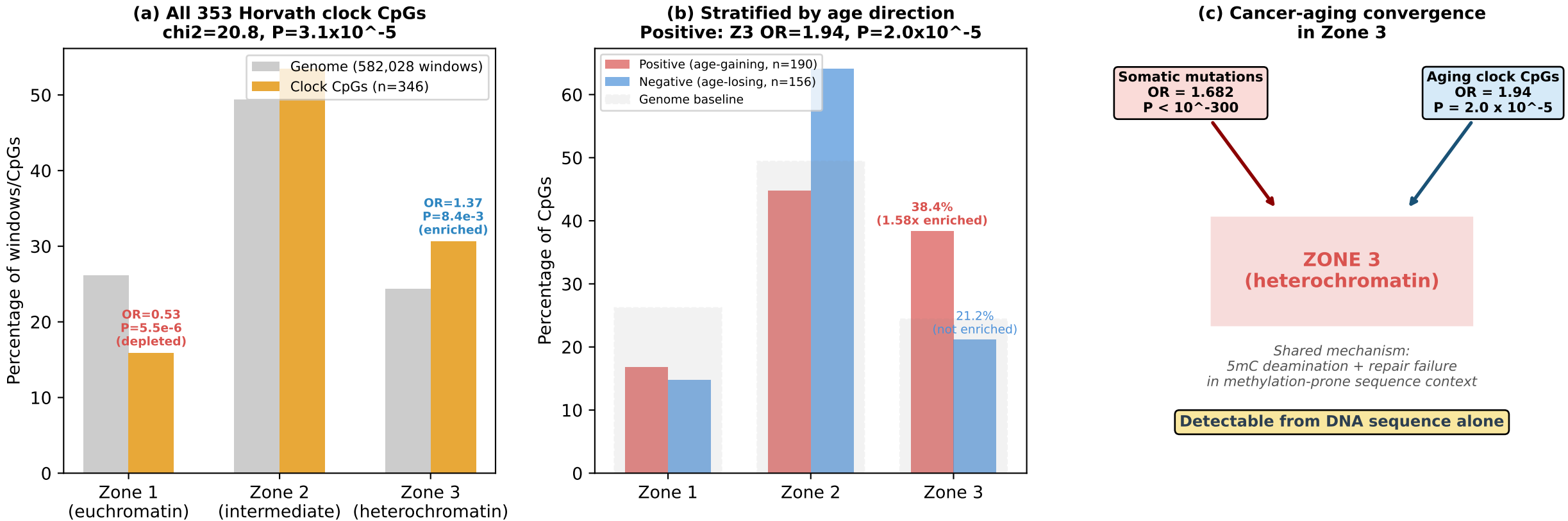

Zone 3 mutation enrichment across 15 TCGA cancer types
PRAD
UCEC
SKCM
OV
HNSC
LIHC
KIRC
STAD
COAD
THCA
BRCA
BLCA
LUAD
GBM
LUSC
Mean OR=1.63
0.8
1.0
1.2
1.4
1.6
1.8
2.0
Odds ratio (Z3 vs Z1)


Mutation class enrichment in Zone 3 (TCGA-BRCA)
Odds ratio (Z3 vs Z1)
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
***
***
***
***
C>T
C>A
C>G
T>A
T>C
T>G
Substitution class


Zone 3 enrichment within RT strata (chr1)
Odds ratio (Z3 vs Z1)
1.5
1.4
1.3
1.2
1.1
1.0
0.9
p=5.5e-05
p=6.8e-04
p=3.8e-02
Early
Mid
Late
Replication timing tertile


Germline variant density
(1000G, 2,504 individuals)
Germline variants per window
145
140
135
130
125
120
115
110
Z3/Z1 = 1.129
p = 1.5×10⁻⁸²
Zone 1
Zone 2
Zone 3


Constitutional vulnerability
amplified in cancer
Z3/Z1 mutation rate ratio
1.8
1.6
1.4
1.2
1.0
0.8
1.38×
Germline
(healthy)
Cancer
(BRCA)


(a) All 353 Horvath clock CpGs
chi2=20.8, P=3.1x10^-5
Genome (582,028 windows)
Clock CpGs (n=346)
Percentage of windows/CpGs
OR=0.53
P=5.5e-6
(depleted)
OR=1.37
P=8.4e-3
(enriched)
Zone 1
(euchromatin)
Zone 2
(intermediate)
Zone 3
(heterochromatin)
(b) Stratified by age direction
Positive: Z3 OR=1.94, P=2.0x10^-5
Positive (age-gaining, n=190)
Negative (age-losing, n=156)
Genome baseline
Percentage of CpGs
38.4%
(1.58x enriched)
21.2%
(not enriched)
Zone 1
Zone 2
Zone 3
(c) Cancer-aging convergence
in Zone 3
Somatic mutations
OR = 1.682
P < 10^-300
Aging clock CpGs
OR = 1.94
P = 2.0 x 10^-5
ZONE 3
(heterochromatin)
Shared mechanism:
5mC deamination + repair failure
in methylation-prone sequence context
Detectable from DNA sequence alone